

Solution Requirements (Functional & Non-functional)

Date	3 October 2026	Team ID / Name	SWTID-2026-6919
Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value	Maximum Marks	4 Marks
Target Platform	ServiceNow Platform (Utah / Washington DC)	Document Version	v1.0 (Final)

1. Functional Requirements

The functional requirements outline the mandatory system capabilities, features, and security workflows designed to implement field-level script-controlled record access control within ServiceNow.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	ACL Script Configuration & Data Security Rule Definition	- Create Scripted Read Access Control List (ACL) rule on target table (u_incident_confidential). - Implement JavaScript condition logic to evaluate field-level privacy flag (current.u_confidential). - Validate active user session roles (gs.hasRole(), gs.getUser()) against record owner/department before granting read access.
FR-2	Database Query Interception (Before Query Business Rule)	- Create a 'Before Query' Business Rule executing on the u_incident_confidential table. - Dynamically inject current.addQuery() conditions to suppress restricted records at database query time. - Eliminate "Security constraints prevent viewing" warning messages in ServiceNow list views.
FR-3	User Role & Departmental Access Control	- Define custom security roles (u_confidential_reader, u_department_lead). - Restrict form field visibility and write capabilities for confidential records based on role membership. - Enforce record-level restriction across list views, related lists, and global search indexes.
FR-4	Audit Logging & Security Access Verification	- Log unauthorized read or write access attempts in ServiceNow security access logs (sys_audit). - Provide Impersonation testing capabilities for System Administrators to verify dynamic role enforcement. - Retain full audit trail of u_confidential field state toggles for compliance review.
FR-5	Reporting & Dashboard Privacy Enforcement	- Apply table and record security constraints automatically to ServiceNow reporting widgets and dashboards. - Ensure data export operations (PDF, Excel, CSV) strictly enforce ACL rules and omit unauthorized field data.

2. Non-functional Requirements

The non-functional requirements specify quality attributes, operational constraints, system security standards, and performance targets for the proposed ServiceNow ACL solution.

NFR No.	Non-Functional Requirement	Description & Standard Target
NFR -1	Usability	Provides a seamless user experience by suppressing unauthorized rows silently in list views and forms, removing intrusive "Security constraints prevent viewing" error banners.
NFR -2	Security	Enforces zero-trust field and record-level authorization executing strictly server-side; client-side scripts, browser manipulation, or API calls cannot bypass security evaluations.
NFR -3	Reliability	Ensures 100% consistent security evaluation across all access vectors including ServiceNow UI Forms, List Views, Global Search, REST/SOAP APIs, and Reporting engines.
NFR -4	Performance	Executes query filtering at the database layer with negligible overhead (<15ms query execution delta), keeping list view rendering times under 250ms for large datasets.
NFR -5	Availability	Operates with high availability matching the ServiceNow cloud platform target uptime (99.99%), employing fail-closed default security logic in case of evaluation errors.
NFR -6	Scalability	Scales horizontally to support hundreds of concurrent fulfillers and thousands of confidential records without degrading database search or ACL evaluation response times.